

AAE 507: Principles of Dynamics

Problem Set 3 & 4

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Out: Sep 11, 2026

PS3 Due: Sep 18 at 11:59 PM ET

PS4 Due: Sep 25 at 11:59 PM ET

(submission at Brightspace>Content>Gradescope)

Consider a rigid satellite bus B with a rigid solar array attached via a hinge. Fig. 1 shows a top-down view of the satellite. A rigid, thin solar-array paddle A is connected to B via a friction-less hinge H on B . Both [Problem Set 3](#) and [Problem Set 4](#) will focus on the planar motion of the satellite in the inertial frame $\mathcal{N} : \{\hat{n}_1, \hat{n}_2\}$.

Relevant quantities and parameters are defined as follows:

- **Bus:** C_B , m_B , and J_{C_B} denote the bus center of mass (CoM), mass, and moment of inertia about C_B , respectively. The bus is a cube of side length $2l$ and of uniform density.
- **Array:** C_A , m_A , and J_{C_A} denote the array CoM, mass, and moment of inertia about C_A , respectively. The $H-C_A$ distance is d .

The positions of C_B and C_A measured from the origin are given by $\mathbf{r}_{C_B}(t) = x_{C_B}(t)\hat{n}_1 + y_{C_B}(t)\hat{n}_2$ and $\mathbf{r}_{C_A}(t) = x_{C_A}(t)\hat{n}_1 + y_{C_A}(t)\hat{n}_2$, respectively.

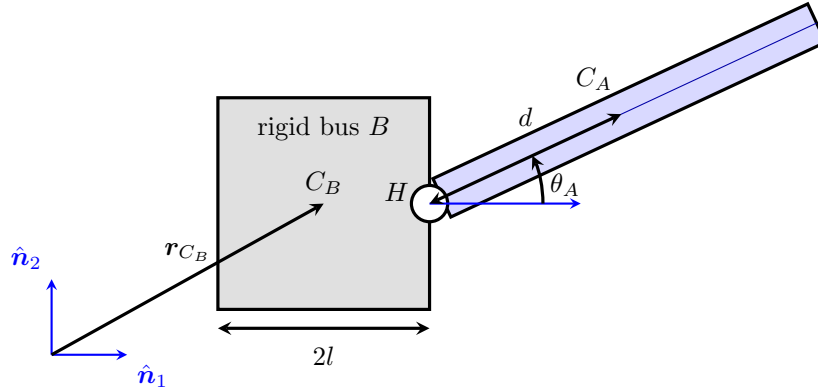


Figure 1: Top-down view of satellite bus B and solar array A , which are both assumed rigid. Both [Problem Set 3](#) and [Problem Set 4](#) consider this system.

Problem Set 3

(3.1) Constraints, Virtual Work, and D'Alembert's Principle

- (a): [**AI use Level: 0**] In this problem, suppose the bus is translationally moving at a constant speed but rotationally fixed in the inertial frame, with the C_B-H line aligned with $\hat{n}_1$.

(a.1): **Discuss** how many degree of freedom (DoF) this system has overall, with justification.

- (a.2): Express the configuration of the system using the Cartesian coordinates $\mathbf{q} = [x_{C_B}, y_{C_B}, x_{C_A}, y_{C_A}]^\top$. The hinge imposes the following velocity-level relation:

$$(x_{C_A} - x_H)\dot{x}_{C_A} + (y_{C_A} - y_H)\dot{y}_{C_A} - (x_{C_A} - x_H)\dot{x}_H - (y_{C_A} - y_H)\dot{y}_H = 0, \quad (1)$$

where $\mathbf{r}_H(t) = x_H(t)\hat{\mathbf{n}}_1 + y_H(t)\hat{\mathbf{n}}_2$ denotes the hinge position. **Identify** the Pfaffian coefficients $a_1(\mathbf{q}, t), \dots, a_4(\mathbf{q}, t)$, and $b(\mathbf{q}, t)$, without using x_H, y_H . **Discuss** how many configuration variables and apparent degrees of freedom this description has.

- (a.3): **Show** that Eq. (1) is integrable using the exactness conditions from class, and **recover** a holonomic EOC, where use the known $H-C_A$ distance to determine the integration constant.
- (a.4): **Discuss** the physical interpretations of both constraint forms (Eq. (1) and the one you obtain in the previous question).
- (a.5): Eliminate the redundant coordinates by solving the obtained holonomic EOC for y_{C_A} and define a minimal set of coordinates $\mathbf{q}' = [x_{C_B}, y_{C_B}, x_{C_A}]^\top$. **Express** $\mathbf{r}_{C_A}(\mathbf{q}', t)$, $d\mathbf{r}_{C_A}$, and $\delta\mathbf{r}_{C_A}$.
- (a.6): Now, consider a different set of minimal coordinates $\mathbf{q}'' = [x_{C_B}, y_{C_B}, \theta_A]^\top$, where θ_A is the array angle measured counterclockwise from $\hat{\mathbf{n}}_1$ to the $H-C_A$ line. **Express** $\mathbf{r}_{C_A}(\mathbf{q}'', t)$, $d\mathbf{r}_{C_A}$, and $\delta\mathbf{r}_{C_A}$. **Discuss** how this approach compares to the one taken in the previous question.
- (b): [**AI use Level: 0**] In this problem, suppose the bus is inertially fixed, and consider the motion of the array relative to the hinge. Assume that the hinge has a torsional spring with potential

$$V_H(\theta_A) = \frac{k_H}{2}(\theta_A - \theta_0)^2 - \tau_H\theta_A, \quad k_H > 0, \quad (2)$$

where θ_0 is the spring reference angle and τ_H is a constant bias torque.

- (b.1): Using the principle of virtual work, **find** every equilibrium $\bar{\theta}_A$.
- (b.2): The hinge constraint force can be nonzero at equilibrium. **Discuss** why this does not contradict its zero virtual work.
- (c): [**AI use Level: 0**] In this problem, suppose the bus is moving in a prescribed way only translationally without rotation, as follows:

$$\mathbf{r}_{C_B}(t) = X_B \cos \Omega t \hat{\mathbf{n}}_1, \quad \mathbf{r}_H(t) = \mathbf{r}_{C_B}(t) + l\hat{\mathbf{n}}_1. \quad (3)$$

For this problem, represent the rigid array A by two particles P_1 and P_2 , each of mass $m_A/2$, connected by a massless rigid rod. Their positions relative to C_A are $\pm\rho_A\hat{\mathbf{e}}_A$, where

$$\hat{\mathbf{e}}_A = \cos \theta_A \hat{\mathbf{n}}_1 + \sin \theta_A \hat{\mathbf{n}}_2, \quad \rho_A = \sqrt{\frac{J_{C_A}}{m_A}}. \quad (4)$$

Thus, this particle system has the same mass and moment of inertia as array A . The torsional spring and bias torque at H are as defined in Eq. (2); no other applied forces act on A .

Denote the positions of P_1, P_2 as $\mathbf{r}_1, \mathbf{r}_2$, respectively, and generalized coordinates as $\mathbf{q} = \theta_A$.

- (c.1): **Express** $\mathbf{r}_i(\mathbf{q}, t)$ and $\delta\mathbf{r}_i$ for $i = 1, 2$.
- (c.2): **Derive** the generalized force $\mathbf{Q}$ associated with the generalized coordinates $\mathbf{q}$.
- (c.3): **Derive** the second time derivative of $\mathbf{r}_i$ kinematically, for $i = 1, 2$.
- (c.4): Using the D'Alembert's principle, **derive** the scalar EOM in θ_A . **Discuss** why the constraint force at H does not appear in the derived EOM.
- (c.5): **Optional (extra credit)**, [**AI use Level: 1**]:
In the derived EOM, **identify** the term induced by the prescribed bus acceleration and **discuss** its physical interpretation.

Problem Set 4

(4.1) Lagrange's Equations

- (a): [**AI use Level: 0**] **Derive** the Lagrange's equations for non-conservative systems with minimal coordinates $\mathbf{q}$, starting from the Lagrangian form of D'Alembert's principle.

Hint: covered in class.

- (b): [**AI use Level: 0**] Consider array A attached to the bus B that is moving in the same manner as in Eq. (3) (translational motion only). The array has a torsional spring at H with potential

$$V_H(\theta_A) = \frac{k_H}{2}(\theta_A - \theta_0)^2 \quad (5)$$

- (b.1): **Derive** the expression of $\mathbf{r}_{C_A}$ in terms of $\mathbf{r}_H, d, \theta_A, \hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2$.

- (b.2): **Calculate** the kinetic energy

$$T_A = \frac{m_A}{2} \dot{\mathbf{r}}_{C_A} \cdot \dot{\mathbf{r}}_{C_A} + \frac{J_{C_A}}{2} \dot{\theta}_A^2 \quad (6)$$

and form $L_A(\theta_A, \dot{\theta}_A, t) = T_A - V_H$.

- (b.3): **Verify** the cancellation-of-dots identity $\partial \dot{\mathbf{r}}_{C_A} / \partial \dot{\theta}_A = \partial \mathbf{r}_{C_A} / \partial \theta_A$.

- (b.4): **Derive** the Lagrange equation for θ_A .

- (b.5): **Discuss** how this EOM compares to the EOM you derived in Problem Set 3(c) with the bias torque τ_H set to zero.

- (c): [**AI use Level: 0**] Suppose the bus is translationally moving as given in Eq. (3) but not rotating (same as the previous problem). Now, let us add the following motor-and-damper torque to the hinge:

$$\tau_H(\dot{\theta}_A, t) = \tau_{H,a}(t) - c_H \dot{\theta}_A, \quad c_H > 0 \quad (7)$$

- (c.1): **Derive** the forced Lagrange equation for θ_A .

- (c.2): Consider commanding the motor at H to produce a prescribed array motion $\theta_A^*(t)$ while bus B follows the given translational motion Eq. (3). **Derive** the required motor torque $\tau_{H,a}(t)$. **Identify** the contributions associated with array inertia, hinge damping, the spring restoring torque, and prescribed bus acceleration.

- (c.3): **Optional (extra credit), [AI use Level: 1]:**

Now, consider a scenario where the satellite is thrusting (= generating translational acceleration) with $\mathbf{a}_H = a_0 \hat{\mathbf{n}}_1$ for a constant a_0 , and the motor is off ($\tau_{H,a} = 0$) in Eq. (7).

Let us analyze the stability of this system. (i) **Determine** the equilibrium array angles $\bar{\theta}_A$.

(ii) **Linearize** the EOM about the determined equilibrium. (iii) **Analyze** its local stability, identifying the condition for the thrust acceleration magnitude a_0 to ensure asymptotic stability.

(4.2) Lagrange Multipliers for Systems with Constraints

- (d): [**AI use Level: 0**] Suppose the bus B is inertially fixed. Use the following coordinates to describe the configuration of A : $\mathbf{q} = [x_{C_A}, y_{C_A}, \theta_A]^\top$. The rigid hinge imposes

$$\psi_1 = x_{C_A} - d \cos \theta_A = 0, \quad \psi_2 = y_{C_A} - d \sin \theta_A = 0. \quad (8)$$

The torsional spring at H has potential $V_H(\theta_A) = \frac{k_H}{2}(\theta_A - \theta_0)^2$. The system is under the applied force at C_A and the applied torque at H , given as $\mathbf{F}_A = F_{x,A} \hat{\mathbf{n}}_1 + F_{y,A} \hat{\mathbf{n}}_2$ and $\tau_{H,a}$, respectively.

- (d.1): **Derive** the Lagrangian $L(\mathbf{q}, \dot{\mathbf{q}}, t)$.

- (d.2): **Derive** the Lagrange equations using multipliers $\lambda_{H,1}$ and $\lambda_{H,2}$.
- (d.3): **Discuss** how the two multipliers are related to the hinge constraint force and its torque about C_A .
- (d.4): **Discuss** why this five-equation formulation (the three Lagrange's equations and the two constraints in Eq. (8)) is equivalent to the one-coordinate expression in Section (4.1), and why the multiplier formulation is useful for hinge-load analysis.
- (d.5): **[AI use Level: 1]**
 At a given instant, let $\theta_A = \theta_0 = 0$, $\dot{\theta}_A = \omega_0$, $\mathbf{F}_A = \mathbf{0}$, and $\tau_{H,a} = \tau_0$. Do not assume $\ddot{\theta}_A = 0$. **Determine** $\ddot{\theta}_A$, the hinge constraint force, and its moment about C_A .
- (e): Now, suppose the bus B is rotating in the inertial frame (no translational motion). In particular, the bus undergoes a prescribed planar slew $\theta_B(t)$.
- (e.1): **[AI use Level: 0]**
Express $\mathbf{r}_{C_A}$ and $\dot{\mathbf{r}}_{C_A}$ as functions of θ_A and $\dot{\theta}_A$. **Discuss** how this changes the contributions to the array angular velocity and CoM velocity. No full EOM derivation is required.
- (e.2): **Optional (extra credit), [AI use Level: 1]:**
Derive the full EOM for this system. Be sure to clearly define relevant quantities and show the derivation steps with justification.